Multi-Lingual Speech Recognition Using A Single Model

IE 643: Deep Learning - Theory and Practice

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Problem Statement & Key Challenges

The Goal

- **One unified Transformer** encoder–decoder shared across all target languages, no per-language models.
- **Native Language Identification (LID)** for every utterance, to be shown live in a demo UI.
- **6 target languages:** English, Hindi, Marathi, Tamil, Bengali + one low-resource language (Maithili / Konkani / Santali).
- **High robustness** to background noise and mix-language speech (Hinglish).

The Challenges

- **Severe data imbalance:** Hindi/English have 1,000s of hours while Bengali/Marathi/Tamil far less and low-resource tier have only 9–175 h in open corpora.
- **LID collapse:** fine-tuning Whisper on low-resource languages destroys its built-in language token (documented failure).
- **Script & metric problem:** Bengali/Malayalam raw WER > 100%; CER vs WER; spelling variation.
- **Hallucinations & noise:** Whisper sometimes invents text on non-speech and denoisers in front of ASR are proven to hurt.

Project goals → Measurable targets

Deliverable	Target / definition of done
Unified ASR model	One checkpoint, shared encoder–decoder, LoRA/QLoRA fine-tune of Whisper large-v3-turbo (809M) or smaller fallback
Integrated LID	Encoder LID-CTC head + supervised decoder language token; ≥ 90% LID accuracy on FLEURS-Indic; confusion matrix reported
Noise robustness	Trained with MUSAN/RIR/SpecAugment; degradation vs clean ≤ agreed margin across SNR sweep; no denoiser front-end
Code-switching	MUCS Hindi-English + synthetic Hinglish in training; CS evaluated separately from monolingual
Demo	Gradio UI: audio in → transcript + language label (+ confidence) out

Why Transformers for Multilingual ASR?

From HMM/GMM → Attention

- Conformer/Transformer encoders model long-range acoustic context **(self-attention)** + local conv structure.
- Encoder–decoder attention gives a **single shared multilingual representation space** — the key enabler for one model across 6 languages.
- Cross-lingual transfer: phoneme inventories overlap heavily across Indic languages (Devanagari family, Dravidian family).

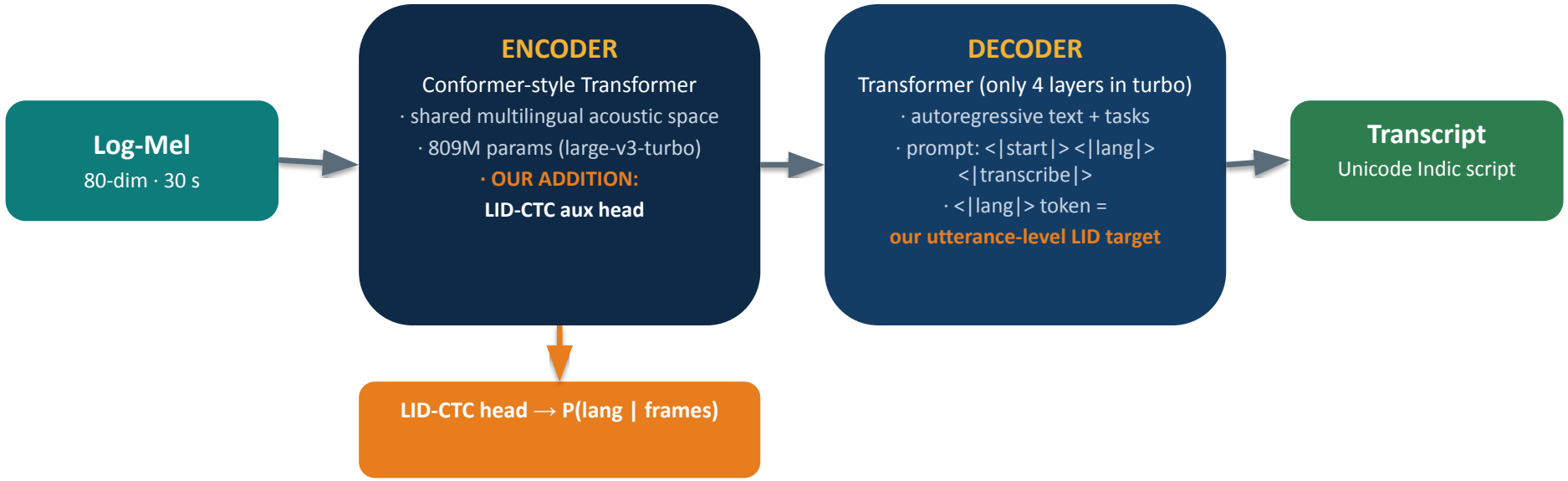
What we read (and what it taught us)

- **Radford et al., Whisper (ICML 2023):** 680k h weakly supervised → robust zero-shot multilingual ASR; 99 languages.
- **Peng et al., OWSM v4 (Interspeech 2025):** fully-open Whisper-style recipe, 166k h / 75 langs; reproducible training pipeline.
- **Qwen3-ASR tech report (2026):** LLM-based ASR does LID+ASR natively — a credible 2026 alternative design.

Three families of multilingual ASR — and where each stands for our task

Family	Examples	Strengths	Weakness for this project
Encoder–decoder (seq2seq)	Whisper large-v3(-turbo), OWSM v4, IndicWhisper	Native language token = built-in LID; strong zero-shot; huge pre-training	Autoregressive hallucinations; LID degrades after fine-tuning
CTC / Transducer (non-autoregressive)	IndicConformer-600M, Parakeet-TDT, OWSM-CTC	Fast, stable decoding; strong on Indic (AI4Bharat)	LID needs an extra head/routing; no built-in token
LLM-based ASR	Qwen3-ASR-0.6B/1.7B, MLC-SLM winners	All-in-one LID+ASR; SOTA open accuracy; code-switch friendly	Heavier inference; less mature fine-tuning recipes

Anatomy of Whisper — Our Starting Point



Key facts

- Trained on 680k h, 99 langs (MIT license)
- turbo: decoder 32→4 layers, ~6× faster, ≈ large-v2 quality
- Known weak spot: Bengali/Malayalam raw WER > 100% → needs fine-tuning + normalization
- Timestamp + translation tokens retained for demo options

Why large-v3-turbo as the base — the accuracy / compute / LID sweet spot

- Half the params of large-v3 (809M vs 1.55B) at ~large-v2 accuracy → fits a student GPU with LoRA/QLoRA.
- Keeps the 99-language <|lang|> token vocabulary — the only strong open base whose LID is **inside** the architecture (unlike CTC models).
- Whisper-small (244M) serves as the fast dev model to validate the full pipeline before committing GPU-hours.
- European 2025 SOTA models (Canary-1B-v2, Parakeet-v3) cover **zero Indian languages** — unusable as a base; we borrow their training tricks instead.

Fine-Tuning a 1B Model on a Student Budget — PEFT Literature

LoRA / QLoRA evidence

- **LoRA-Whisper (Song et al., Interspeech 2024):** multilingual WER **11.68%** → **9.52%** (monolingual oracle 9.19%) at ~5% trainable params; per-language LoRA matrices mitigate interference.
- **HF PEFT INT8/QLoRA recipe:** Whisper-large fine-tunes in < 8 GB VRAM (free T4); Marathi 14.01 vs 13.64 full-FT WER — near-parity.
- **Layer freezing (CKA analysis):** bottom ~⅓ of encoder layers are language-agnostic; freezing them cuts WER 4–6% relative and curbs overfitting.

Multilingual & low-resource tricks

- **Weighted cross-entropy (Piñeiro-Martín et al., 2024):** up-weight rare-language tokens in one joint fine-tune; LID learned "for free" by keeping <|lang|> in labels.
- **O-LoRA / rehearsal-free (Xu et al., 2024):** add a new language without catastrophic forgetting — needed for our low-resource tier.
- **Task arithmetic (Rafkin et al., 2026):** merge per-language task vectors with a high-resource "support" language — training-free boost for Maithili/Konkani/Santali.

Our training configuration (chosen from this literature)

Component	Choice	Why (source)
Dev model	Whisper-small + QLoRA, 4-bit	Pipeline validation on free T4/Kaggle before committing GPU (HF PEFT recipe)
Final model	Whisper large-v3-turbo (809M) + LoRA r=16–32	~6× faster than large-v3 at ≈large-v2 accuracy (OpenAI model card); ~5% trainable params (LoRA-Whisper)
Frozen layers	Bottom ~⅓ of encoder	Language-agnostic layers; +4–6% rel. WER, less overfitting (CKA study)
Loss	ASR CE (+ token weighting) + λ · LID-CTC (tuned)	Joint LID+ASR prevents LID collapse (Piñeiro-Martín; LID-CTC paper)
Optimizer	AdamW, LR 1e-5 (5e-5 for new params), batch 16	Standard in Whisper FT literature (Interspeech 2024–2026)

Language ID Integrated with ASR — the Core Research Problem

The failure we must design against

- **Documented collapse:** After fine-tuning Whisper on low-resource languages, its built-in LID token silently degrades (OpenAI Whisper Discussion #1454; multiple papers).
- **Root cause:** The model overfits the dominant language(s) of the fine-tune mix and stops attending to the language token.
- **Consequence:** Wrong <|lang|> ⟶ wrong script/vocabulary at decode time ⟶ cascading transcription errors.
- **Fixes that work:** (a)Ttrain <|lang|> as a supervised target; (b) add an explicit auxiliary encoder LID head; (c) balance the mix.

Evidence that the auxiliary head pays off

- **LID-CTC on MMS-1B (Wang et al., Interspeech 2025, ML-SUPERB 2.0, 2nd place):** +14% relative LID accuracy, **−30% relative CER** (⟶ 82.4% LID / 16.3% CER).
- **Joint DID+ASR, Indian languages (Kumar et al., Interspeech 2025):** 81.63% dialect-ID, 4.65% CER across 8 Indian languages / 33 dialects; **detach the ID gradient** from ASR and never prepend ID tokens to text.
- **Routing cost is real:** TalTech ML-SUPERB winner: oracle vs predicted LID = ~4 CER points ⟶ we will report both.

Our hybrid LID design

Encoder LID-CTC head	Decoder <|lang|> token	Demo LID display	External baseline
Frame-level P(lang|audio); multi-task loss λ·LID-CTC; detached from ASR branch (Kumar et al.)	Utterance-level LID, kept as supervised target during fine-tune (prevents collapse)	Head's softmax ⟶ label + confidence in Gradio UI; confusion matrix in report	SpeechBrain ECAPA-TDNN (VoxLingua107, 107 langs) for comparison

Data Balancing, Noise Robustness & Hallucination Control

Imbalanced tiers → balancing strategy

- **Polyglot-Lion (Dang & Ngo, MeLLM 2026):** two-stage deterministic upsampling — (1) equalize corpora within a language, (2) equal share across languages; beat temperature sampling, exact parity guaranteed. We keep language tags (they drop them).
- **Canary-1B-v2 (NVIDIA):** cosine-annealed sampling weights toward uniform during fine-tune.
- **Our choice:** Polyglot-Lion upsampling to an equal per-language share across the 6 languages, with a cap on any single corpus's dominance.

Noise: do NOT put a denoiser in front

- **When De-noising Hurts (2025):** MetricGAN+ degraded ASR in **all 40 configs** (Whisper, Parakeet, Gemini × 10 conditions); noisy beat enhanced everywhere.
- **SAM-Audio + Whisper (2026):** denoising improves perceptual quality but consistently raises WER/CER — worse for larger models.
- **What works instead:** VAD segmentation + training-time augmentation (MUSAN noise, RIR reverb, SpecAugment, speed perturb).

Hallucination control — teach the model when NOT to speak

- **Calm-Whisper (Wang et al., Interspeech 2025):** just 3 of 20 decoder self-attention heads cause >75% of non-speech hallucinations; fine-tuning those heads on non-speech audio with blank labels → **>80% hallucination cut, <0.1% WER cost**.
- **Canary recipe:** 36,000 h of non-speech audio paired with empty-string targets during training.
- **paza-whisper post-processing:** words-per-second truncation + repetition detection kills trailing hallucinations at inference without touching weights.
- **Our plan:** Silero VAD for segmentation only · MUSAN/RIR/SpecAugment augmentation · non-speech blank-label data · optional WPS truncation. No denoiser front-end, ever.

Existing open-source Multilingual ASR Models

Model (year)	Params	Type	Indian langs?	LID	Key numbers / verdict
Whisper large-v3 (2023)	1.55B	Enc-Dec	Weak (Bn/Ml >100% raw WER)	Token	680k h pretrain; strong Hindi, broken Bengali/Malayalam
Whisper large-v3-turbo (2024)	809M	Enc-Dec (4 dec. layers)	Weak as-is → fine-tune	Token	≈large-v2 quality, ~6× faster → OUR BASE
Whisper-small (2023)	244M	Enc-Dec	Weak	Token	FLEURS Hi 86.9% / Ta 93.3% raw → DEV MODEL
OWSM v4 (2025)	~1B	Enc-Dec	Limited	Yes	166k h/75 langs; fully reproducible ESPnet recipe
Qwen3-ASR-0.6B / 1.7B (2026)	0.6/1.7B	LLM-ASR	52 langs+dialects (incl. Indic)	Built-in	SOTA open ASR; TTFT 92 ms; stretch-goal baseline
Canary-1B-v2 (2025)	978M	FastConformer + Dec	None (25 EU only)	Auto	Methods reference: balancing + anti-hallucination tricks
Parakeet-TDT-0.6b-v3 (2025)	600M	Conformer+TDT	None (25 EU only)	Auto	Fast; not usable for Indic
MMS-1B-all (2023)	1B	w2v2 + adapters	1000+ (many Indic)	Yes	ML-SUPERB 2nd with LoRA: 82.4% LID / 16.3% CER
SeamlessM4T-v2 (2023)	2.3B	Enc-Dec	Strong Indic	Yes	FLEURS: Marathi 10.7 vs Whisper 24.3 CER

Takeaway: Most global models don’t really cover much Indian languages. For our Indic baseline, we have narrowed down to the Whisper-family, Qwen3-ASR, and AI4Bharat models.

Existing open-source Multilingual ASR Models

Model	Params	Coverage	License	Role in our project
AI4Bharat IndicConformer-600M	600M	All 22 scheduled languages (CTC+RNNT)	MIT	PRIMARY Indic baseline to beat/compare
AI4Bharat IndicWhisper	~769M (medium)	12 Indian langs, Vistaar fine-tune	MIT	Reference: Whisper fine-tuned on Indic data
SpeechBrain ECAPA LID (VoxLingua107)	~20M	107 languages	Apache 2.0	External LID-only baseline for our LID head
Microsoft paza-whisper-turbo	809M	Low-resource (Kenyan) langs	MIT	Reuse WPS truncation trick at inference

Our plan to use these models:

- **IndicConformer-600M:** we run it on our benchmark suite to produce the per-language baseline table (its 22-lang coverage makes it the fairest single comparison).
- **IndicWhisper:** proof that Whisper + Indic fine-tuning data works (best WER on 39/59 Vistaar benchmarks, avg −4.1 WER) — our approach extends it with LID supervision + balancing.
- **Qwen3-ASR-0.6B:** 2026 LLM-ASR challenger with built-in 52-lang LID; Polyglot-Lion already shows a balanced Tamil fine-tune — stretch goal: reproduce it as a second model.
- **Canary / Parakeet:** not usable as bases (no Indian languages) — we adopt their data-balancing and 36k-h non-speech anti-hallucination recipes.

What Our Base Model Was Originally Trained On

Whisper pre-training corpus (Radford et al., 2023)

- **680,000 hours** of weakly labeled audio–transcript pairs scraped from the web (MIT license)
- **117,000 h (17%) non-English** across 99 languages, Indic coverage exists but is thin and noisy
- **Multi-task format in one model:** transcription, translation, language ID, timestamps, VAD via special tokens (SOT, <|lang|>, <|transcribe|>, <|translate|>)
- **Why relevant to us:** the multilingual backbone + token-based LID is exactly the shared-representation starting point our task requires

Known gaps we must close with fine-tuning

- **Indic scripts under-trained:** Bengali & Malayalam raw WER > 100%, Hindi 22.0 WER (best of Indic)
- **Domain skew:** web audio ≠ Indian telephonic/spontaneous speech, accent & code-switch under-represented
- **LID brittleness:** token LID is the weakest link on low-resource languages, and degrades further under fine-tuning
- **All three gaps are addressed by our plan:** Indic fine-tune data + LID-CTC (Connectionist Temporal Classification) head + balanced sampling + augmentation

Pre-training scale vs our fine-tune scale (low compute budget)

	Whisper pre-training	Our fine-tuning (planned)
Data	680k h weakly labeled web audio	300–500 h curated, transcribed, balanced across 6 languages
Compute	Unpublished (industrial scale)	Single 24GB GPU (LoRA/QLoRA)
Objective	Multi-task seq2seq on 99 langs	ASR CE + λ·LID-CTC multi-task, 6 languages

Fine-Tuning Data We Will Use (Collection & Curation)

Corpus	Langs	Hours	Role for our 6-language mix
IndicVoices (AI4Bharat)	22 Indian	23.7k total / 11.2k transcribed	Primary source incl. our low-resource tier (spontaneous, real speakers)
Vistaar-train (AI4Bharat)	12 Indian	10.7k h	Core fine-tune pool (Shrutilipi, NPTEL, IndicTTS)
Shrutilipi (AIR news)	12 Indian	6,457 h (Hi 1,620, Mr 1,015, Ta 794, Bn 443)	Read-speech backbone; largest clean per-language hours
MUCS 2021 (MSR)	7 Indian + Hinglish/Bn-En pairs	600 h	Code-switch training + evaluation
Kathbath / IndicSUPERB	12 Indian	1,684 h	Held-out evaluation + LID benchmark
Common Voice 22 + OpenSLR	Hi/Mr/Ta/Bn	Bn 1,276 h / Ta 226 h / Mr 109 h	Supplementary read speech (Bengali is unusually strong here)

Curation pipeline:

Dedup & filter (drop >30 s clips, bad transcripts) → Unicode NFC + lowercase + punctuation strip → Whisper+IndicNLP text normalization → two-stage balancing upsampling to equal per-language share → split by speaker (no leakage) → augmentation: MUSAN/RIR/SpecAugment/speed-perturb + non-speech blank-label clips.

Evaluation Benchmarks & Metrics

Benchmark	Focus	Size	Why in our suite
FLEURS (Indic subset)	14 Indian langs	12 h/lang	Standard multilingual test set, raw Whisper is broken here
Vistaar (59 benchmarks)	12 langs × domains	varies	Indic-standard, IndicWhisper best on 39/59 → direct comparison
IndicVoices test	22 langs, spontaneous	held-out	Real conversational Indian speech
Kathbath / IndicSUPERB	12 langs + LID task	from 1,684 h corpus	Includes a dedicated Indic LID evaluation
Svarah + Lahaja	Indian-accented English, multi-accent Hindi	9.6 h / multi	Accent robustness for English & Hindi tiers
MUCS code-switch	Hinglish, Bn-En	test sets	Isolated CS evaluation (reported separately)

Metric protocol (adopted from ML-SUPERB 2.0)

- **WER and CER** (CER primary for Tamil/Bengali, agglutinative scripts, spelling variants)
- **Normalizers:** Whisper BasicTextNormalizer + IndicNLP, Unicode NFC, report raw AND normalized
- **Aggregate stats:** mean, worst performing languages, etc.

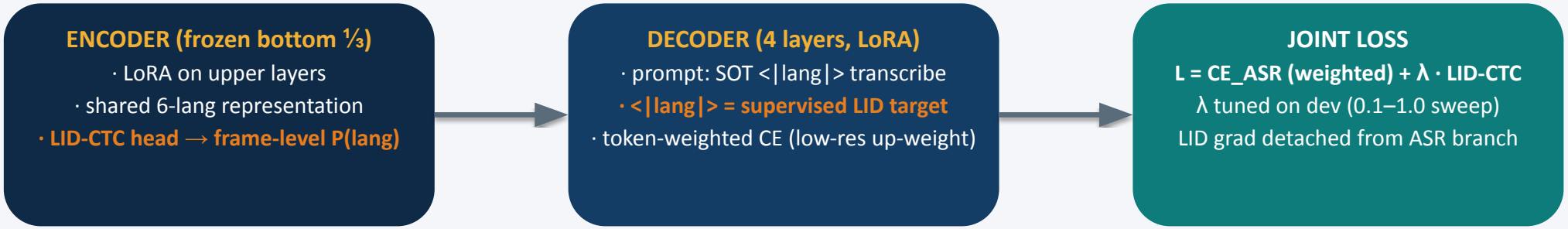
LID-specific reporting

- **LID accuracy + 6×6 confusion matrix** on IndicSUPERB / FLEURS
- **Oracle-vs-predicted-LID ASR gap** — quantifies routing cost (≈4 CER points in published systems)
- **Qualitative error typology:** substitution / deletion / insertion / hallucination / normalization errors, with examples

End-to-End Pipeline

1 · INPUT AUDIO	2 · PREPROCESS	3 · TRAINING MIX	4 · MODEL	5 · OUTPUT
Any 6-language clip Hinglish CS · noisy · telephonic	Silero VAD segment 16 kHz · NFC text side NO denoiser	Two-stage balanced upsampling equal per-language share + MUSAN/RIR/SpecAugment + non-speech blank clips	Whisper large-v3-turbo + LoRA adapters (r=16–32) + encoder LID-CTC head λ·LID-CTC + token-weighted CE	Transcript (Unicode script) + LID label & confidence + optional WPS anti-hallucination

Inside the model — multi-task training (single loss, two LID signals)



Why two LID signals?

Frame-level head = robust, trainable supervision.

Token = free utterance-level LID + drives decoding.

Evidence: +14% LID / −30% CER (IS'25).

Handling Code-Switching (Hinglish)

What the literature tells us

- **MUCS 2021 (Diwan et al.):** the standard Hindi-English / Bengali-English CS challenge, baseline WER 32% on CS sub-task - hard, and that is with 600 h
- **CS transfer is limited:** Paik et al. (2026) show model merging gives only modest transfer of bilingual CS ability to unseen pairs - naive merging is insufficient
- **What helps:** CS-specific training data - MUCS corpora + synthetic TTS code-mixing (IITG-HingCoS sentences) + transliteration/dual-script decoding.

Our CS strategy

- **Train:** mix MUCS Hindi-English (+ optionally Bengali-English) into the balanced 6-language pool, augment with synthetic Hinglish from TTS
- **Decode:** script-aware - Devanagari vs Latin output decided by predicted language per segment, no forced single-script hypothesis
- **Evaluate:** CS reported separately from monolingual WER, expectation-setting stated up front (we do not promise unseen-pair zero-shot CS)

Code-switch expectation

Scenario	Expectation	Evidence
Seen pair (Hindi–English), MUCS-style	Significant improvement over vanilla Whisper, close to monolingual-tier WER	IndicWhisper/Vistaar gains, CS-specific data works
Synthetic Hinglish (TTS-augmented)	Better function-word switching accuracy, measured on IITG-HingCoS test	Code-mixing synthesis literature
Unseen CS pairs (e.g., Tamil–English at inference)	Modest at best - explicitly out of scope for claims	Paik et al. 2026: merging gives only modest transfer

Noise Robustness

Rejected: speech-enhancement front-end

- **MetricGAN+ / SAM-Audio / DeepFilterNet-style denoisers** degraded ASR in all tested configs (2025–2026 studies, 4 ASR systems × 10 noise conditions).
- **Even on clean audio, denoising adds a 1.3–3.2% WER penalty** - perceptual quality ≠ recognition accuracy.
- **Decision: no denoiser in the pipeline**

Adopted: training-time + inference-time levers

- **Augmentation:** MUSAN noise, RIR reverberation, SpecAugment, speed perturbation (±10%).
- **Anti-hallucination data:** non-speech clips (AudioSet/DEMAND/UrbanSound) with blank-string targets (Calm-Whisper/Canary recipe).
- **Inference guardrails:** Silero VAD segmentation, optional words-per-second truncation + repetition detection (paza-whisper).

Robustness evaluation protocol

Test condition	Construction	Metric reported
Clean (baseline)	Unmodified benchmark test sets	WER/CER per language
SNR sweep	Mix MUSAN/DEMAND noise at 20 / 15 / 10 / 5 / 0 dB into FLEURS + IndicVoices test	WER/CER vs SNR curve per language
Reverberant	RIR-convolved subset	WER/CER degradation vs clean
Non-speech robustness	UrbanSound-style clips through the model	Hallucination rate: % outputs with >5 tokens on non-speech (target: >80% reduction vs vanilla)
Long-form / telephonic	Voice of India-style long clips via VAD chunking	WER/CER + segmentation-boundary error

Experiments & Investigations Carried Out So Far

Completed work (prep phase)

- **Literature survey** of several papers (2023–2026) on multilingual ASR, LID, PEFT, balancing, and Indic corpora - basis of this deck
- **Baseline scouting:** ran IndicConformer-600M inference on sample clips across the 6 target languages to sanity-check coverage and outputs (AI4Bharat NeMo fork)
- **Data audit:** downloaded & profiled IndicVoices + Vistaar (per-language hours, transcript quality, script mix)

Preliminary observations

- **Confirms literature:** vanilla Whisper-small gives raw FLEURS WER of 86.9% (Hindi) / 93.3% (Tamil) - garbage before normalization, IndicConformer outputs are clean but with no usable LID signal for our demo
- **Normalization matters:** applying Whisper+IndicNLP normalizer moves Hindi score by 22 points, Tamil by 41 points absolute - metric protocol must be fixed first
- **Data reality:** low-resource tier (Maithili/Konkani/Santali) confirmed thin in open corpora - task-arithmetic/LoRA-warm-start fallback validated as necessary

Next experiments

#	Experiment	Decision it informs
E1	LoRA vs QLoRA vs full-FT on Whisper-small, Hindi-only, fixed steps	Which PEFT config to scale up
E2	LID-CTC head on/off (λ sweep 0.1–1.0), 3-language mix	λ value; does LID supervision hurt ASR?
E3	Balanced vs unbalanced sampling, 3-language mix	Quantify balancing gain on worst language
E4	SNR sweep on fine-tuned vs vanilla Whisper-small	Does augmentation recipe suffice without denoiser?

FUTURE PLAN

Execution Roadmap

STAGE 1	**Pipeline & data foundation** Lock metric/normalizer protocol · finalize 6-language mix & curation · E1–E4 on Whisper-small · reproducible training/eval repo
STAGE 2	**Scale-up run** LoRA fine-tune Whisper large-v3-turbo on balanced mix · tune λ for LID-CTC · add non-speech blank data · first full benchmark table
STAGE 3	**Robustness & CS** MUSAN/RIR/SpecAugment schedule · SNR sweep eval · MUCS Hinglish + synthetic CS training · CS evaluation suite
STAGE 4	**Low-resource tier & ablations** LoRA warm-start / task-arithmetic for Maithili/Konkani/Santali · ablation study (LID head, balancing, augmentation) · error analysis
STAGE 5	**Demo, baselines & report** Gradio demo (transcript + LID + confidence) · full IndicConformer/Qwen3-ASR baseline tables · confusion matrices · final report + video

FUTURE PLAN

Risks and Fallbacks

Risk / trigger	Fallback action
large-v3-turbo LoRA exceeds compute/latency budget	Drop to Whisper-small or a CTC head (Parakeet-style)
LID accuracy < 90% on FLEURS-Indic	Increase λ (LID loss weight), strengthen encoder LID-CTC head
Mid-resource language WER stuck > 40%	Add IndicVoices/Shrutilipi hours or a language-specific LoRA adapter
Code-switch WER unacceptable	Add synthetic Hinglish (TTS) + dual-script decoding
Low-resource tier (Maithili/Konkani/Santali) overfits	LoRA warm-start from nearest mid-resource language + task-arithmetic merge

FUTURE PLAN

Expected Outcomes & Deliverables

Unified checkpoint

One Whisper large-v3-turbo + LoRA model, 6 languages, shared encoder–decoder and no per-language variants

Integrated LID system

Encoder LID-CTC head + supervised <|lang|> token, LID confusion matrix and oracle-vs-predicted-LID gap reported

Baseline comparison tables

Per-language WER/CER against IndicConformer-600M and Qwen3-ASR-0.6B, on every benchmark in our suite

Robustness results

SNR-sweep WER/CER curves plus a hallucination-rate reduction on non-speech audio (target: >80% vs vanilla Whisper)

Working demo

Gradio UI: audio in → transcript + language label + confidence out, running on a single 24GB GPU

Ablation studies

LID head on/off · balanced vs unbalanced sampling · augmentation on/off

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Summary

- **Base model** — Whisper large-v3-turbo + LoRA (Whisper-small QLoRA for the dev pipeline)
- **LID design** — Encoder LID-CTC head + supervised <|lang|> token — evidence: +14% relative LID accuracy, −30% relative CER
- **Data strategy** — Two-stage balanced upsampling across IndicVoices, Vistaar, Shrutilipi and MUCS
- **Robustness** — No denoiser front-end; MUSAN/RIR/SpecAugment + blank-label non-speech data + VAD + WPS guard

Thank you